Day – 2 Introduction to VPNs Theory.

Task – 1 Create to separate networks.

- Step 1: Setting up the Network Topology in Packet Tracer.
- To simulate two separate branch offices, build the following layout in your workspace:
- Branch A Office: Place one Cisco router (e.g., Router 1 / 2911), one switch (Switch 1 / 2960), and one end device (PC 0). Connect PC 0 to Switch 1 using a copper straight-through cable, and connect Switch 1 to Router 1's GigabitEthernet interface.
- Branch B Office: Place a second router (Router 2), switch (Switch 2), and end device (PC 1). Connect them similarly with straight-through cables.
- Step 2: Defining the IP Addressing Scheme
- Assign separate subnet ranges to each branch so they function as distinct local networks:
- Branch A LAN: 192.168.1.0/24
- Router 1 Interface (GigabitEthernet 0/0): 192.168.1.1 (Subnet Mask: 255.255.255.0)
- PC 0 IP Address: 192.168.1.2 (Subnet Mask: 255.255.255.0, Default Gateway: 192.168.1.1)
- Branch B LAN: 192.168.2.0/24
- Router 2 Interface (GigabitEthernet 0/0): 192.168.2.1 (Subnet Mask: 255.255.255.0)
- PC 1 IP Address: 192.168.2.2 (Subnet Mask: 255.255.255.0, Default Gateway: 192.168.2.1)
- Step 3: Configuring Router 1 (Branch A)
- Click on Router 1, go to the CLI tab, and enter the following commands to configure its local interface:
- Plaintext
- Router> enable
- Router# configure terminal
- Router(config)# hostname RouterA
- RouterA(config)# interface gigabitEthernet 0/0
- RouterA(config-if)# ip address 192.168.1.1 255.255.255.0
- RouterA(config-if)# no shutdown
- RouterA(config-if)# exit
- RouterA(config)# exit
- RouterA# write memory

- Step 4: Configuring Router 2 (Branch B)
- Click on Router 2, go to the CLI tab, and execute the configuration commands for its interface:
- Plaintext
- Router> enable
- Router# configure terminal
- Router(config)# hostname RouterB
- RouterB(config)# interface gigabitEthernet 0/0
- RouterB(config-if)# ip address 192.168.2.1 255.255.255.0
- RouterB(config-if)# no shutdown
- RouterB(config-if)# exit
- RouterB(config)# exit
- RouterB# write memory
- Step 5: Configuring the End Devices (PCs)
- Configure PC 0 (Branch A):
- Click PC 0 -> Desktop tab -> IP Configuration.
- Set IPv4 Address to 192.168.1.2, Subnet Mask to 255.255.255.0, and Default Gateway to 192.168.1.1.
- Configure PC 1 (Branch B):
- Click PC 1 -> Desktop tab -> IP Configuration.
- Set IPv4 Address to 192.168.2.2, Subnet Mask to 255.255.255.0, and Default Gateway to 192.168.2.1.

Task – 2 Config a basic site to site VPN Betwwen two routers.

- To establish a secure and direct connection between the two branch offices over an unsecure public WAN link, you can configure a GRE (Generic Routing Encapsulation) tunnel.
- Assuming you have already configured the LAN interfaces (GigabitEthernet 0/0) as outlined previously, follow these steps to connect Router A and Router B using a WAN link and a GRE tunnel.
- Step 1: Connecting the Routers (WAN Link)
- Connect GigabitEthernet 0/1 on Router A to GigabitEthernet 0/1 on Router B using a Copper Cross-Over cable (or via a simulated cloud/switch representing the ISP).
- Assign an interconnecting public subnet (e.g., 10.0.0.0/30) to these interfaces so the routers can talk to each other across the WAN.
- Step 2: Configuring the WAN Interfaces
- On Router A:
- RouterA> enable

- RouterA# configure terminal
- RouterA(config)# interface gigabitEthernet 0/1
- RouterA(config-if)# ip address 10.0.0.1 255.255.255.252
- RouterA(config-if)# no shutdown
- RouterA(config-if)# exit
- On Router B:
- RouterB> enable
- RouterB# configure terminal
- RouterB(config)# interface gigabitEthernet 0/1
- RouterB(config-if)# ip address 10.0.0.2 255.255.255.252
- RouterB(config-if)# no shutdown
- RouterB(config-if)# exit
- Step 3: Configuring the GRE Tunnel on Router A
- Define a tunnel interface (Tunnel 0), assign a tunnel IP address (192.168.100.1/30), and point the source and destination to the physical WAN IP addresses.
- RouterA(config)# interface tunnel 0
- RouterA(config-if)# ip address 192.168.100.1 255.255.255.252
- RouterA(config-if)# tunnel source gigabitEthernet 0/1
- RouterA(config-if)# tunnel destination 10.0.0.2
- RouterA(config-if)# exit
- Step 4: Configuring the GRE Tunnel on Router B
- Mirror the setup on Router B, using its local WAN interface as the source and Router A's WAN IP as the destination, with the corresponding tunnel IP (192.168.100.2/30).
- RouterB(config)# interface tunnel 0
- RouterB(config-if)# ip address 192.168.100.2 255.255.255.252
- RouterB(config-if)# tunnel source gigabitEthernet 0/1
- RouterB(config-if)# tunnel destination 10.0.0.1
- RouterB(config-if)# exit
- Step 5: Adding Static Routes for Branch Traffic
- For the PCs to reach the opposite branch, the routers need to know that traffic destined for the remote LAN must be sent through the GRE tunnel.
- On Router A (to reach Branch B's LAN 192.168.2.0/24):
- RouterA(config)# ip route 192.168.2.0 255.255.255.0 tunnel 0
- RouterA(config)# exit
- RouterA# write memory
- On Router B (to reach Branch A's LAN 192.168.1.0/24):
- RouterB(config)# ip route 192.168.1.0 255.255.255.0 tunnel 0
- RouterB(config)# exit
- RouterB# write memory

Task – 3 Simulate a remote access VPN scenario.

- Simulating a remote access VPN in Cisco Packet Tracer allows an individual remote worker (like a mobile user or employee working from home) to securely connect back to a branch office network.
- Here is how to set up and configure a remote access VPN connection using Packet Tracer's built-in PC VPN client feature (typically supported when connecting a PC to a VPN-capable router like the Linksys WRT300N or a configured branch gateway).
- Step 1: Setting Up the VPN Server (Router/Gateway)
- To accept remote VPN connections, the receiving router or wireless gateway must have VPN server settings enabled:
- Click on the branch router or wireless router acting as your VPN gateway.
- Go to the GUI tab (if using a wireless router model like WRT300N) or configure the IPsec/PPTP server settings via CLI.
- Enable the VPN Server feature.
- Create user credentials for the remote worker:
- Username: remote_user
- Password: SecurePass123
- Note down the public-facing IP address or gateway IP of this router (e.g., 10.0.0.2 or the WAN interface IP).
- Step 2: Configuring the VPN Client on the PC
- Packet Tracer's simulation PCs include a dedicated VPN application tool on the desktop interface:
- Click on the Remote PC (the client device simulating the traveling employee) in your workspace.
- Navigate to the Desktop tab.
- Look for and open the VPN application icon.
- Fill in the connection details provided by the server configuration:
- Group Name / Connection Name: Branch-VPN
- Server IP Address: Enter the public IP of the branch router (e.g., 10.0.0.2).
- Username: remote_user
- Password: SecurePass123
- Click Save to store the profile.
- Step 3: Establishing the Secure Tunnel
- In the same VPN application window on the PC, select your saved profile (Branch-VPN).
- Click the Connect button.
- Watch the interface status change; a successful handshake will authenticate the credentials and establish an encrypted virtual tunnel interface over the simulated public cloud.

Task – 4 Differences between site to site VPN and remote access VPN.

- Main Differences Between Site-to-Site VPN and Remote Access VPN
- Scope and Target Audience: A site-to-site VPN connects entire networks to each other (such as connecting an entire branch office network to a headquarters data center), allowing all devices behind one router to communicate transparently with devices behind another. A remote access VPN connects an individual user's device (like a traveling employee's laptop or mobile phone) securely to a private corporate network.
- Architecture and Connection Method: In a site-to-site setup (configured via GRE tunnels between routers), the routers themselves handle the encapsulation and routing continuously in the background, requiring no special software on individual end devices. In a remote access setup, an individual client explicitly runs a VPN client application, logs in with credentials, and establishes an on-demand tunnel from their remote device.
- User Involvement: Site-to-site connections are always-on and completely transparent to end-users (PCs in Branch A just ping PCs in Branch B normally). Remote access requires user action, manual login authentication, and active session initiation before access is granted.
- Site-to-Site VPN
- Advantages
- Seamless, always-on connectivity: Once configured on the routers, users do not need to log in or configure any software; all traffic between offices flows securely and automatically.
- Centralized management: Security and routing policies are managed entirely at the gateway routers, protecting all connected devices within the branch simultaneously.
- Disadvantages
- High initial setup complexity: Requires dedicated routers, specific static configurations, and synchronized tunnel parameters on both ends.
- Fixed location dependency: It only connects pre-defined physical locations; it cannot accommodate dynamic or traveling workers out of the box.
- Remote Access VPN
- Advantages
- High flexibility and mobility: Empowers remote, field, or work-from-home employees to securely access corporate applications and servers from anywhere in the world.
- Cost-effective for individual users: Eliminates the need for hardware routers at every remote worker's home, requiring only client software on existing devices.
- Disadvantages

- User friction: Requires individual users to manually remember credentials, launch the client app, and initiate connections before accessing resources.
- Single-device vulnerability: If an individual remote worker's device is compromised, it can potentially act as an entry point into the broader corporate network unless strict endpoint security is enforced.

Task – 5 Step by step config commands for site to site VPN.

- Router A configuration
- 1. WAN interface configuration
- RouterA> enable
- RouterA# configure terminal
- RouterA(config)# interface gigabitEthernet 0/1
- RouterA(config-if)# ip address 10.0.0.1 255.255.255.252
- RouterA(config-if)# no shutdown
- RouterA(config-if)# exit
- 2. GRE tunnel interface configuration
- RouterA(config)# interface tunnel 0
- RouterA(config-if)# ip address 192.168.100.1 255.255.255.252
- RouterA(config-if)# tunnel source gigabitEthernet 0/1
- RouterA(config-if)# tunnel destination 10.0.0.2
- RouterA(config-if)# exit
- 3. Static routing configuration
- RouterA(config)# ip route 192.168.2.0 255.255.255.0 tunnel 0
- RouterA(config)# exit
- RouterA# write memory
- Router B configuration
- 1. WAN interface configuration
- RouterB> enable
- RouterB# configure terminal
- RouterB(config)# interface gigabitEthernet 0/1
- RouterB(config-if)# ip address 10.0.0.2 255.255.255.252
- RouterB(config-if)# no shutdown
- RouterB(config-if)# exit
- 2. GRE tunnel interface configuration
- RouterB(config)# interface tunnel 0
- RouterB(config-if)# ip address 192.168.100.2 255.255.255.252
- RouterB(config-if)# tunnel source gigabitEthernet 0/1
- RouterB(config-if)# tunnel destination 10.0.0.1
- RouterB(config-if)# exit

- 3. Static routing configuration
- Plaintext
- RouterB(config)# ip route 192.168.1.0 255.255.255.0 tunnel 0
- RouterB(config)# exit
- RouterB# write memory
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